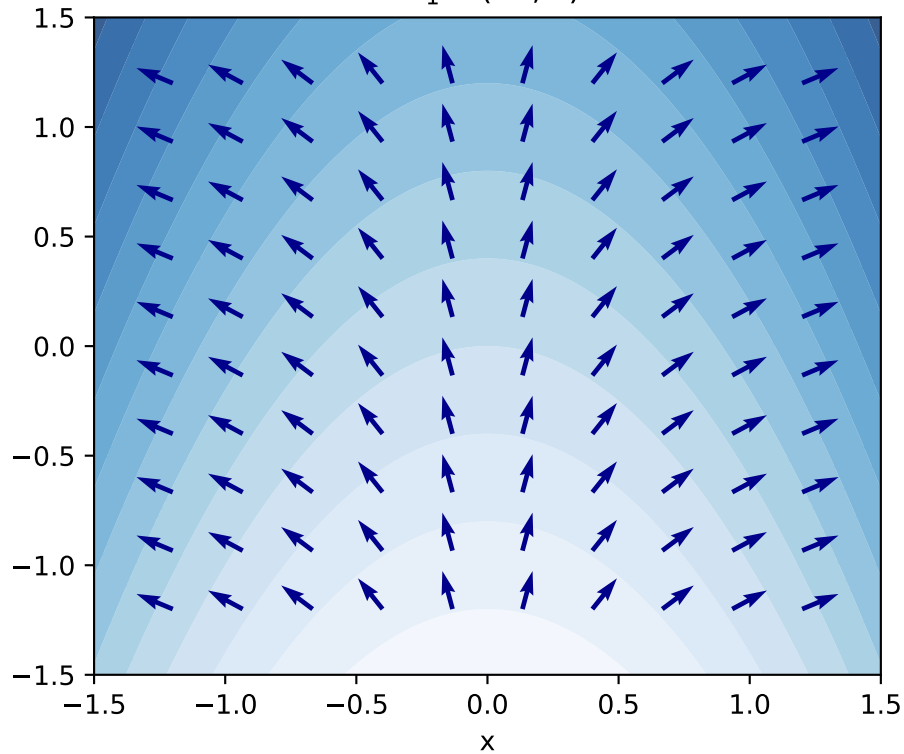
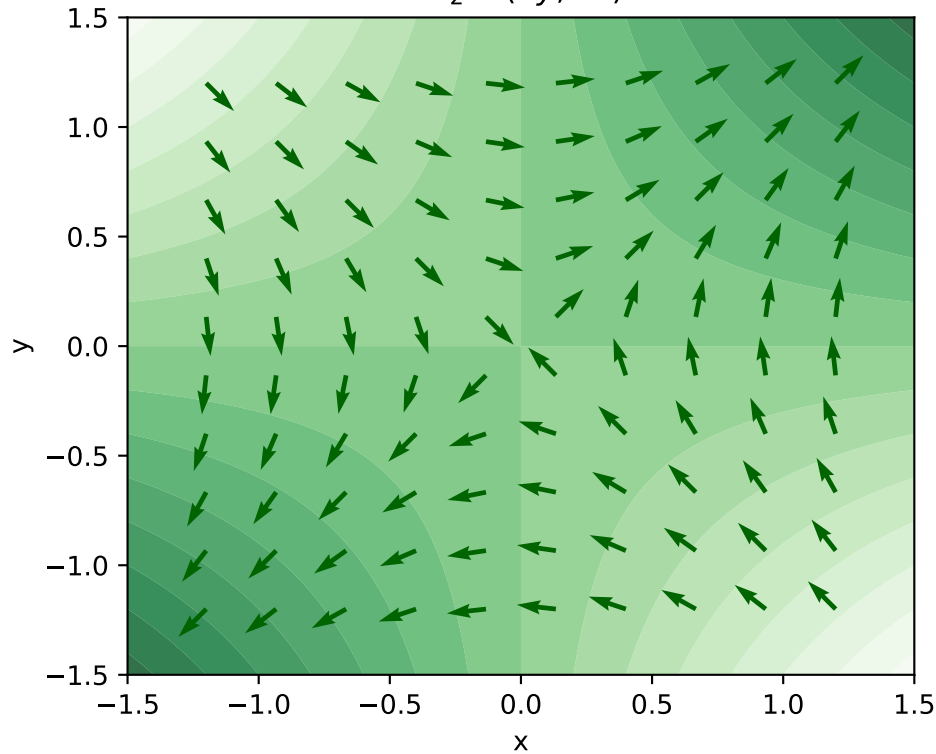


Plot 4: Vektorwertige Funktion - Jacobi-Matrix enthält alle Gradienten

$$f_1(x, y) = x^2 + y$$
$$\nabla f_1 = (2x, 1)$$



$$f_2(x, y) = 2xy$$
$$\nabla f_2 = (2y, 2x)$$



Jacobi-Matrix:

$$f(x, y) = (x^2 + y, 2xy)^T$$

$$J_f = \begin{bmatrix} 2x & 1 \\ 2y & 2x \end{bmatrix}$$

Row 1 = $\text{grad}(f_1)^T$

Row 2 = $\text{grad}(f_2)^T$

BEIDE Gradienten
in einer Matrix!

At (1,1):

$$J = \begin{bmatrix} 2 & 1 \\ 2 & 2 \end{bmatrix}$$